

Nathaniel Brazen

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Education

Colorado School of Mines, Golden, CO

MS, Quantum Engineering (Software Focus)

Expected May 2027

Focus: Quantum Algorithms & Complexity Theory, AI/ML, Computer Architecture & HPC

BS, Computer Science

May 2026

GPA: 3.5/4.0

Experience

Graduate Assistant, Computer Vision (CSCI 437/507) - Colorado School of Mines

Jun 2026 - Present

- Grade and audit coursework in classical/geometric computer vision - camera models and projection matrices, edge detection, geometric image transforms, and 2D/3D scene segmentation
- Provide root-cause debugging and code-review support to peers troubleshooting image-processing algorithms
- Designed a CNN training assignment (ConvNetJS on CIFAR-10) requiring students to compare SGD vs. adaptive optimizers and tune architecture depth, activation choice, and learning rate against validation accuracy

DECTech Summer Intern - Lead Instructor, Colorado School of Mines

May 2026 - Jul 2026

- Designed and personally validated an autonomous soccer framework for VEX AIM robots in Python - applying real-time computer vision object detection to locate the ball and goal, then navigating and executing a scoring kick
- Led a cross-functional team of interns, architecting a Python curriculum from foundational syntax through Gemini GenAI API integration

Project Advisor - Computing For Good, Colorado School of Mines

Jun 2026 - Jul 2026

- Directed a team of developers through Agile/Scrum sprints building a nonprofit tree-carbon calculation database, meeting all goals on schedule
- Advised on a custom neural network modeling tree growth from USDA Forest Inventory data, reviewing modeling approach and data validation methodology

Skills

Languages: Python, C++, Java, C#, JavaScript, Rust

Computer Vision & ML: OpenCV, PyTorch, TensorFlow/Keras, Scikit-learn, Qiskit, classical/geometric CV (camera models, projection, object detection, segmentation)

Systems & Tools: Linux/UNIX (Bash, POSIX threads), Docker, Git/GitHub, GCP, REST APIs, SQL/PostgreSQL, Neo4j

Practices: OOP, data structures & algorithms, unit testing (JUnit, pytest), Agile/Scrum

Projects

Shor's Algorithm Simulator - Python, NumPy, pytest

- Built a from-scratch, dependency-light quantum state-vector simulator (every gate an explicit matrix/tensor operation) implementing the full Shor's algorithm pipeline: n-qubit QFT, phase estimation, and a modular-exponentiation oracle, factoring $N = 15$ end-to-end
- Verified via 217 passing pytest tests covering gate unitarity, QFT-vs-classical-DFT agreement, and dense-vs-tensor-reshape equivalence

Hardware-Aware Circuit Transpiler - Python, Qiskit, NetworkX

- Hand-wrote a qubit-routing pass mapping a `qiskit.QuantumCircuit` onto a fixed hardware topology (2D grid, ring, heavy-hex), inserting BFS-shortest-path SWAP gates to make every 2-qubit gate legal
- Benchmarked the greedy router against a naive baseline on a 5-qubit QFT circuit (20 vs. 23 SWAPs); verified via pytest for adjacency correctness and statevector semantics-preservation

Live-Streaming Platform for Rugby Matches - WebRTC, SRS, Node/Express, Docker, Terraform, GCP

- Built sub-second-latency live video: phones publish via WebRTC (WHIP) to an SRS server, viewers watch in-browser (WHEP); publishing is gated by a Node/Express server with signed session cookies and rate-limited login
- Deployed with Docker Compose, Caddy, and Terraform on a GCP free-tier VM; a static IP as the WebRTC candidate lets phones on carrier NAT connect without TURN

Disaster Tweet Classifier - Python, TensorFlow/Keras, Scikit-learn

- Managed the end-to-end ML lifecycle for 10,000+ social media entries: preprocessing pipeline (Regex + NLTK) and a TensorFlow/Keras Sequential model (Dense/Dropout, Adam optimizer) reaching 0.79 F1
- Benchmarked Logistic Regression, Naive Bayes, SVMs, and neural networks to identify the most scalable real-time approach